

VELVET ESPRESSO CAFÉ

Executive Summary:

Group 4 – Process Pioneers

Emily Cuberos

Vindhya Mahitha Mallampati

Hugo Fajardo

Cinthia Lopez

Carlos Molina

ISM 3153 – Business Process Analysis

Velvet Espresso Café Process Improvement Initiative

Velvet Espresso Café is a café that focuses on providing high-quality coffee, food, and customer service in a welcoming environment. The café's goal is to give customers a fast and enjoyable experience while maintaining accurate orders and efficient operations. As customer expectations continue to increase, the business has identified an opportunity to improve its customer ordering process by using artificial intelligence (AI). This project focuses on analyzing the current ordering process, identifying its weaknesses, and proposing an AI-driven solution that can improve efficiency, customer satisfaction, and overall business performance.

As-Is Analysis

The process selected for improvement is customer order processing. This process was chosen because it directly affects customer wait times, order accuracy, and overall satisfaction. It is also one of the most important processes in the café because it connects customers, employees, inventory, and payment systems.

To better understand the process, a simulated discovery phase was conducted using observations, stakeholder interviews, surveys, and AI-generated personas. During a simulated observation of 30 customer orders during peak hours, the average wait time was 13 minutes. Out of the 30 orders, four required corrections or remakes, resulting in a 13% error rate. In addition, three customers left before placing an order, which represented 10% of the customers observed. The observation also found that 60% of the orders included at least one customization, making the ordering process more complicated.

The stakeholder analysis identified several common concerns. Customers wanted faster service, more accurate orders, and better information about wait times. Cashiers reported that entering customized orders during busy periods often caused mistakes. Baristas explained that unclear instructions frequently led to order remakes. Managers stated that they struggled with predicting demand, staffing levels, and inventory needs during busy times.

Overall, the current process works well when business is slow, but several problems appear during peak hours. Long lines, order errors, and unclear communication create delays that negatively impact both customers and employees.

To-Be Design and AI Solution

To improve the process, the team proposes implementing AI technologies in phases rather than introducing all of them at once. This approach allows the café to measure results, collect customer feedback, and make adjustments before adding additional AI features.

The first phase of implementation will focus on AI voice ordering. Customers will be able to place orders using voice commands instead of manually selecting menu options. This feature will be especially useful for drive-through customers and customers who are in a hurry. AI voice ordering is expected to reduce ordering time, improve convenience, and help employees process orders more efficiently. To prevent issues with complex orders, customers will also have access to a comment box where they can provide special instructions if needed.

After the AI voice ordering system has been implemented and evaluated, the café plans to introduce additional AI enhancements. One future improvement is an AI-powered predictive wait time system that analyzes order volume and staffing levels to provide more accurate pickup estimates. Another future enhancement is an AI chatbot that can answer common customer questions regarding menu items, store hours, promotions, and order status. These additional features will be implemented over time based on performance data and customer feedback.

The redesigned process allows customers to place orders through traditional online ordering or AI voice ordering. Customers can provide special requests through the comment box while employees receive clearer order information. As additional AI features are introduced in later phases, customers will also benefit from more accurate wait times and automated customer support. These improvements are expected to create a faster, more accurate, and more convenient ordering experience.



Technology Justification

The proposed AI technologies were selected because they directly address the issues identified during the discovery phase. Research on AI implementation showed that predictive analytics and AI-powered customer service tools have successfully improved efficiency in many organizations.

For example, Stord uses AI to improve delivery-time predictions, while Gojek uses AI-powered voice assistants to help customers complete tasks more efficiently. These examples demonstrate how AI can improve communication, reduce delays, and increase customer satisfaction.

The proposed solutions are a good fit for Velvet Espresso Café because they improve the current process without requiring major changes to the business. Instead of replacing employees, the technology helps employees perform their jobs more efficiently while improving the customer experience.

There are some potential risks, including voice-recognition errors, inaccurate wait-time predictions, and chatbot limitations. However, these risks can be reduced through regular monitoring, employee oversight, and continuous updates to the AI systems.

Business Impact and KPIs

The proposed AI improvements are expected to create measurable benefits for both customers and employees. To evaluate success, the café will track several Key Performance Indicators (KPIs).

The first KPI is Average Order Processing Time. The current average processing time is approximately 13 minutes, and the goal is to reduce that time by 35–40%.

The second KPI is Order Accuracy Rate. The current process experiences a noticeable number of order corrections and remakes. The goal is to increase order accuracy to at least 98%.

The third KPI is Customer Satisfaction Score (CSAT). Customer satisfaction will be measured through Qualtrics surveys after purchases, with a target score of 4.8 out of 5 or higher.

Additional metrics such as AI adoption rate, repeat customer usage, and customer feedback will also be reviewed regularly. These measurements will help management determine whether the new process is achieving the desired results.

Although AI voice ordering will be implemented first, the café expects additional improvements in efficiency and customer satisfaction as predictive wait times and chatbot capabilities are introduced in future phases. This phased approach reduces implementation risk while allowing management to evaluate performance and make improvements throughout the rollout process.

Overall, the proposed AI-driven improvements are expected to reduce wait times, improve order accuracy, increase customer satisfaction, and create a more efficient ordering process. By implementing these technologies, Velvet Espresso Café can improve its operations while providing customers with a faster and more convenient experience.